

Aviation Accidents



CS-513 Knowledge Discovery & Data Mining Final Project

Presented by

Shahzaib Chaudhary

Michael Moonan

Owen Krupa

Problem Statement

Fatal aviation accidents, while rare, have severe consequences.

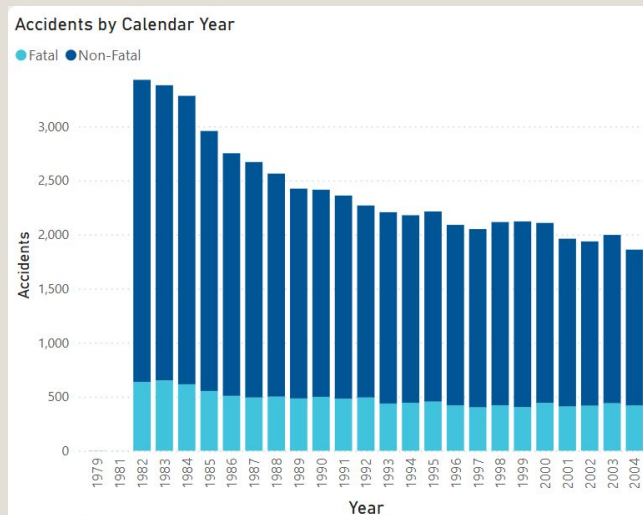
The Good news:

Fatal aviation risk has steadily declined over the course of aviation history, as well as the number of total accidents

The Bad news:

Aviation accidents still happen

58 fatal crashes in 2025 alone



Source:

<https://www.nts.gov/safety/data/Pages/monthly-dashboard.aspx>

Project Goal

1. Identify patterns and risk factors for fatal aviation accidents to support risk assessment, safety audits, and policy planning
2. Provide predictive models that assess likelihood of an aviation accident causing fatalities
3. Advance our understanding of aviation incidents

Data Collection & Initial Analysis

Data obtained from National Travel Safety Board:
<https://www.nts.gov/>

Used past 5 years' data to ensure data is up to date with current aviation standards.

Notes:

- Data is likely human-entered on a system that has been updated, since many incomplete records follow a similar pattern of which attributes are omitted
- After removing bad records, every record still contained at least 1 missing attribute
- Target variable, HighestInjuryLevel, was missing from 57% of records

EDA

Initial Dataset:

8430 records
46 Features

Dataset After Cleaning:

8413 records
22 Features

(Features above 40% null were
automatically excluded)

Feature Construction:

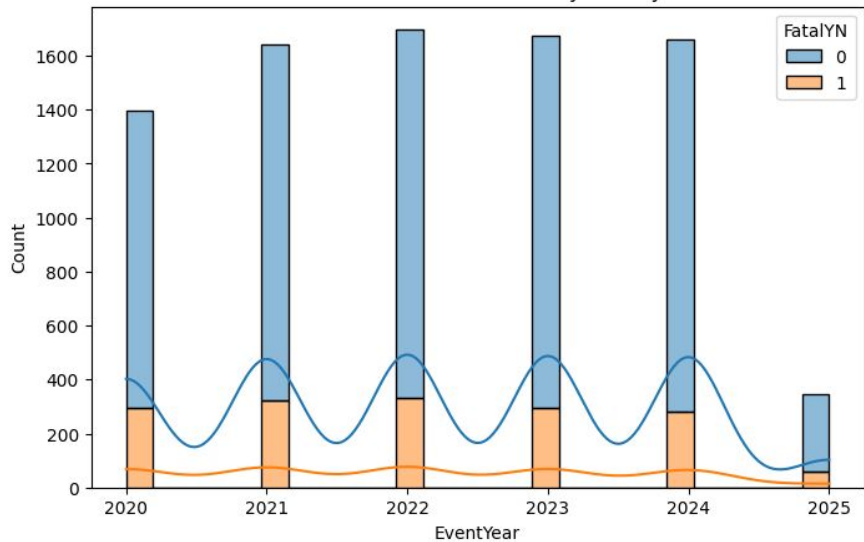
DaysUntilPublished - Used to evaluate NTSB's claim to publish reports within 24 months of an incident.

FatalYN - Target (0=Non-fatal, 1=Fatal)
Calculated via HighestInjuryLevel and FatalInjuryCount

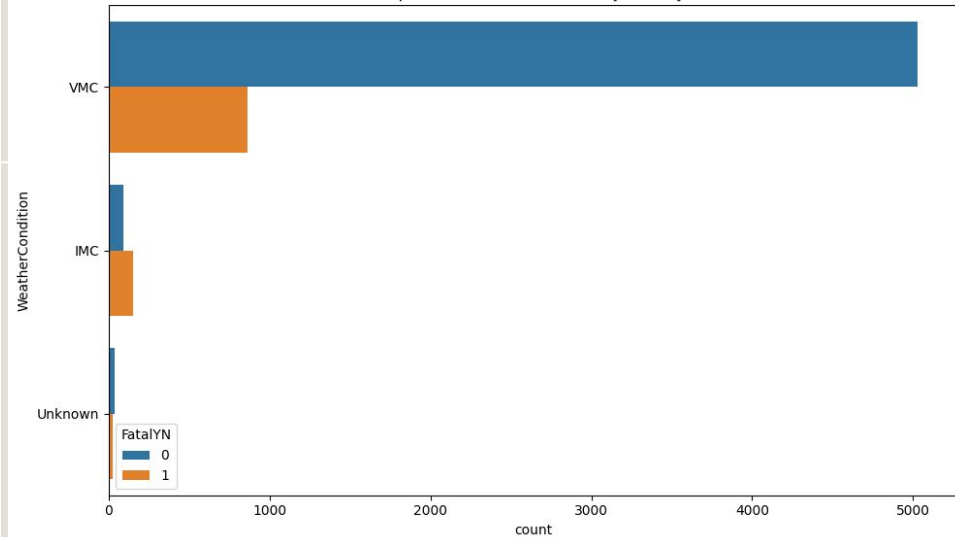
After calculation, distribution was roughly 80% non-fatal, 20% fatal.

We decided to impute missing values and used SMOTE to infer fatal crashes in order to balance the dataset.

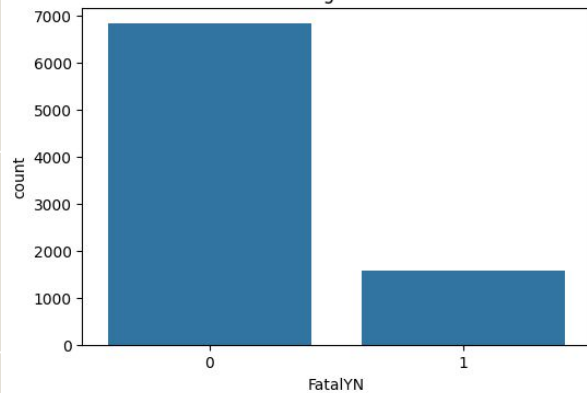
Distribution of EventYear by Fatality



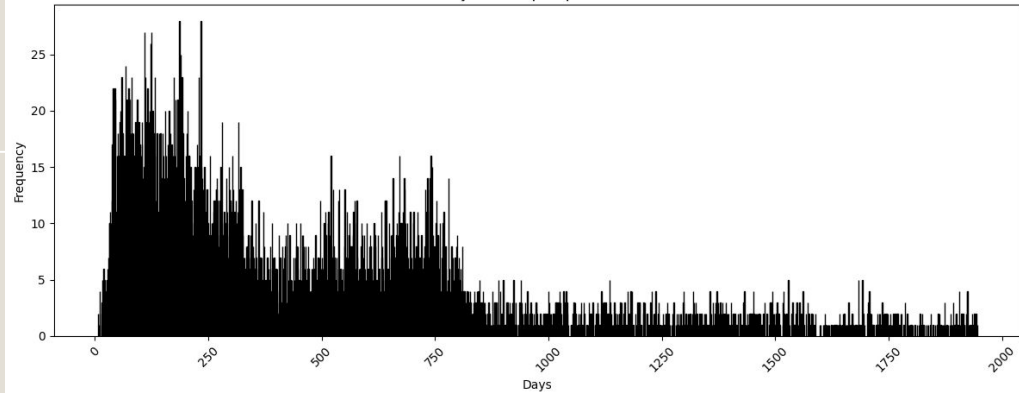
Top 10 Weather Conditions by Fatality



Distribution of Target Variable: FatalYN



Days until report published

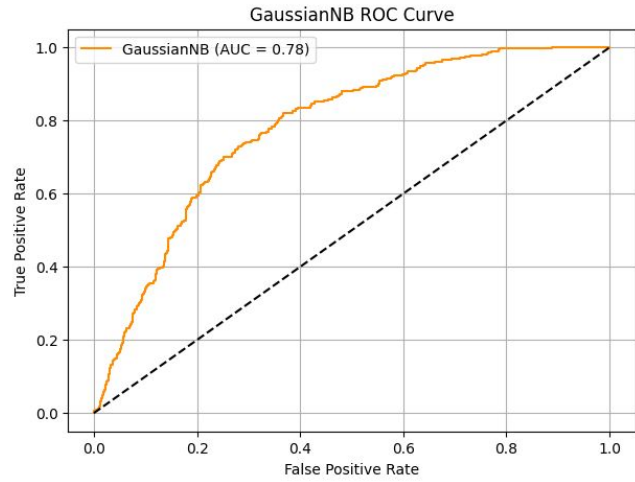
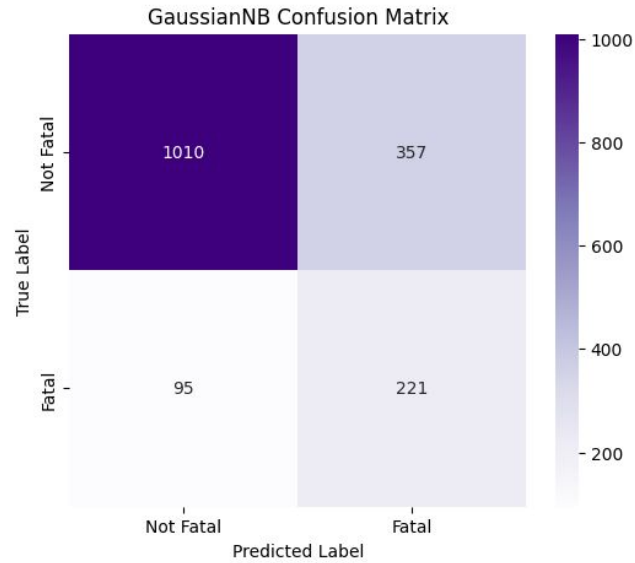
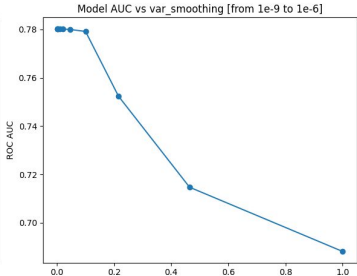
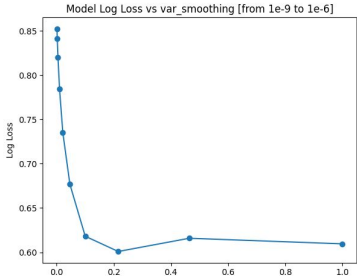
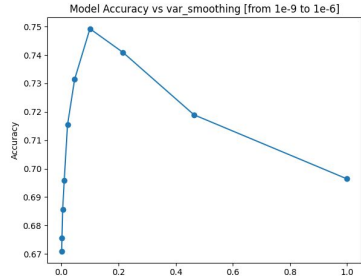


Model: Naive Bayes

Results

GaussianNB Classification Report:

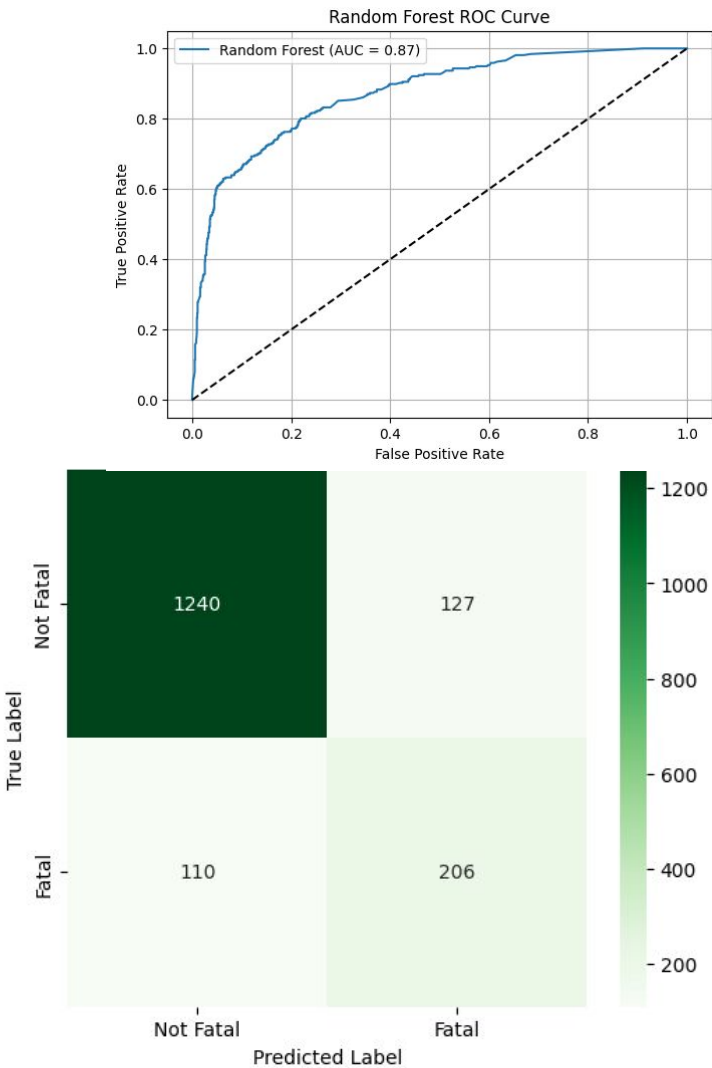
	precision	recall	f1-score	support
Not Fatal (0)	0.91	0.74	0.82	1367
Fatal (1)	0.38	0.70	0.49	316
accuracy			0.73	1683
macro avg	0.65	0.72	0.66	1683
weighted avg	0.81	0.73	0.76	1683



Model: Random Forest

Random Forest Classification Report:

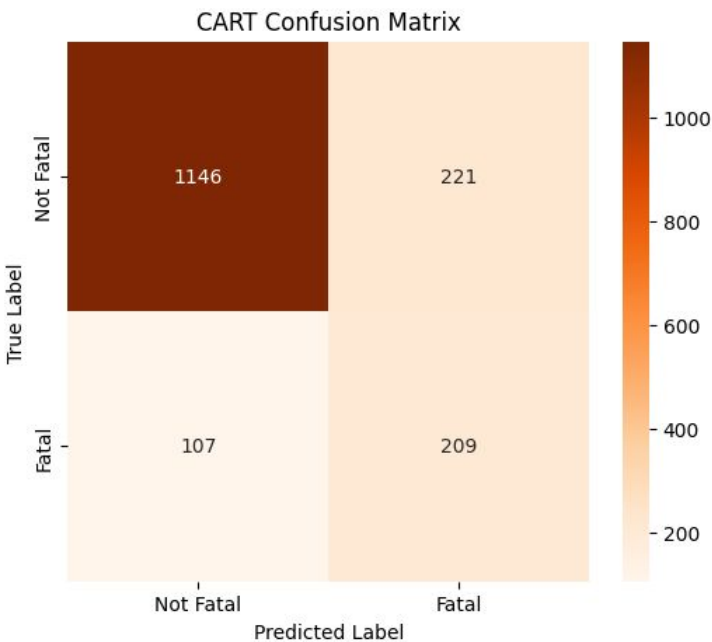
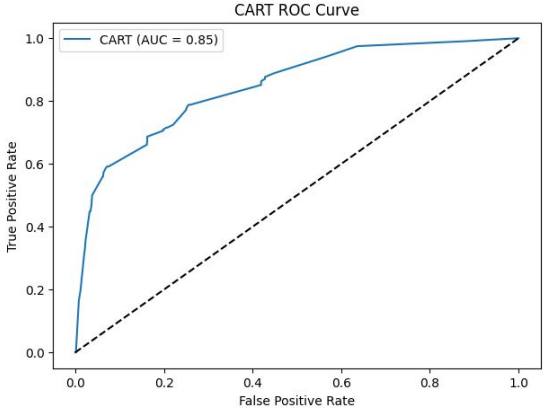
	precision	recall	f1-score	support
Not Fatal (0)	0.92	0.91	0.91	1367
Fatal (1)	0.62	0.65	0.63	316
accuracy			0.86	1683
macro avg	0.77	0.78	0.77	1683
weighted avg	0.86	0.86	0.86	1683



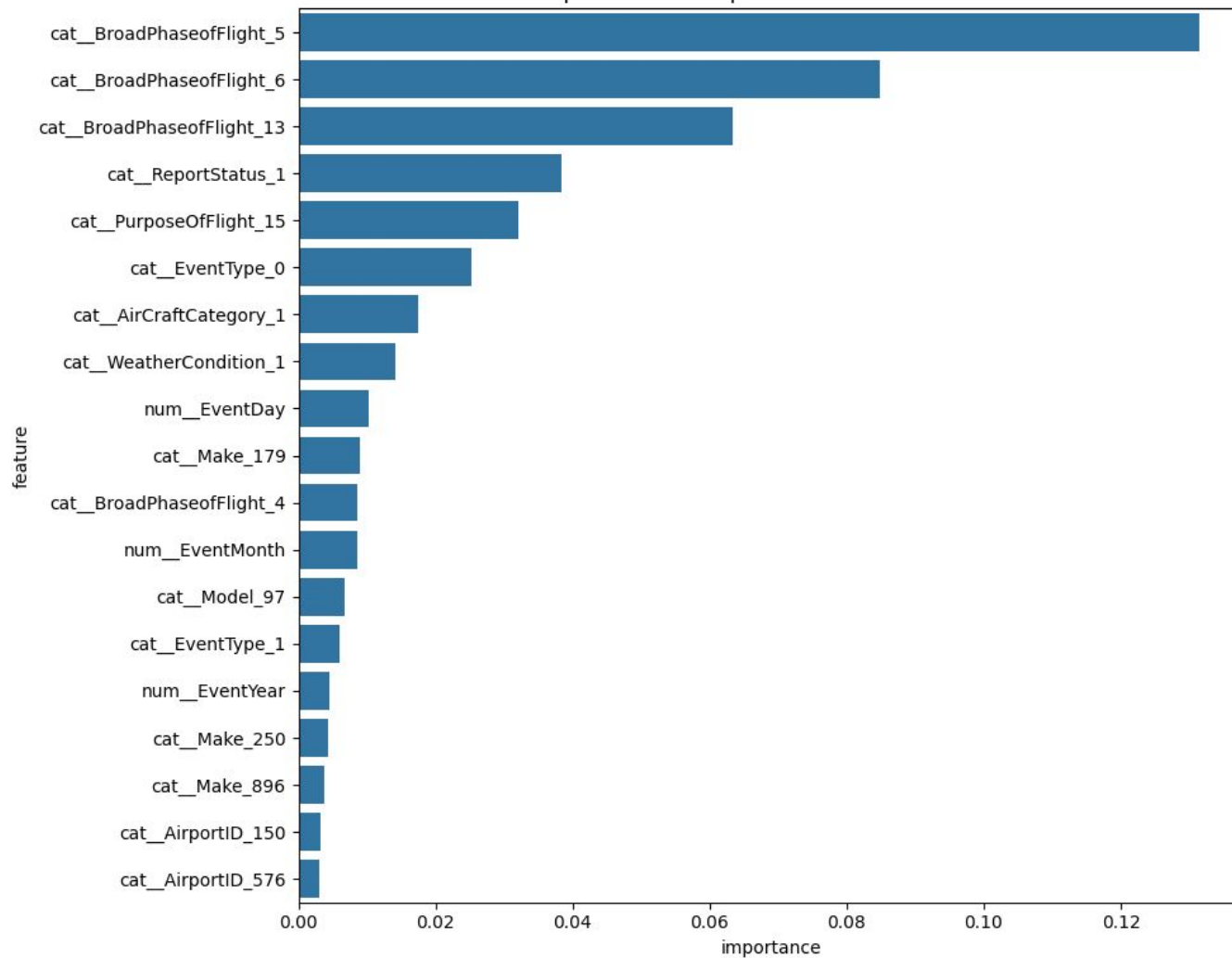
Model: CART

CART Classification Report:

	precision	recall	f1-score	support
Not Fatal (0)	0.91	0.84	0.87	1367
Fatal (1)	0.49	0.66	0.56	316
accuracy			0.81	1683
macro avg	0.70	0.75	0.72	1683
weighted avg	0.83	0.81	0.82	1683

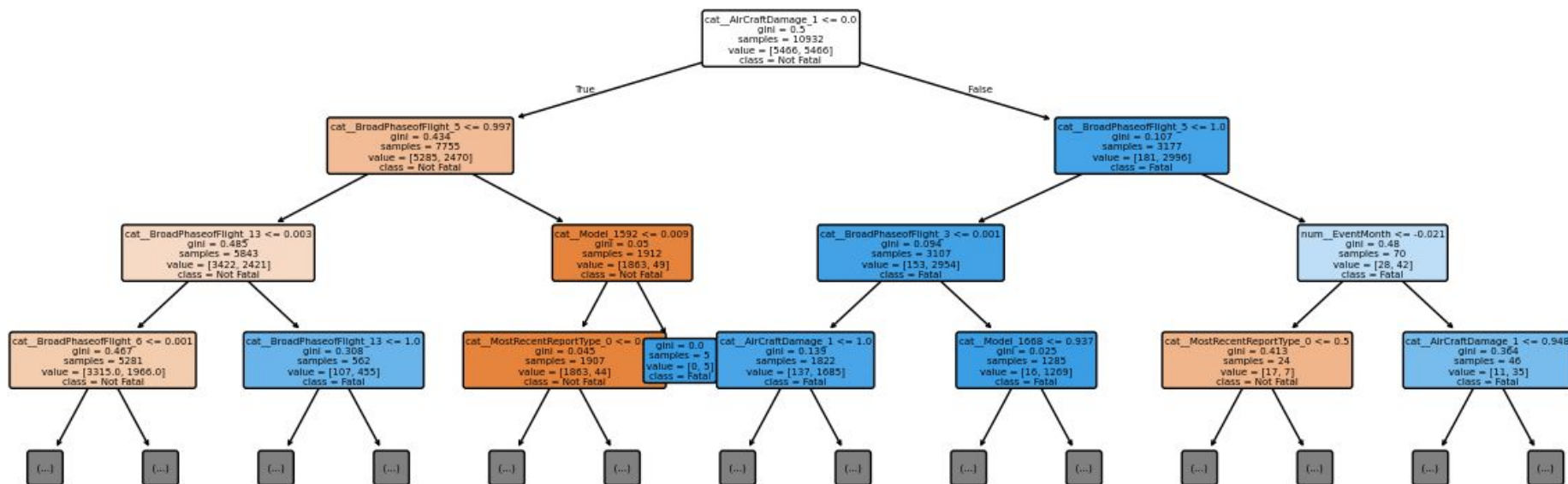


Top 20 Feature Importances for CART Model



Model: CART

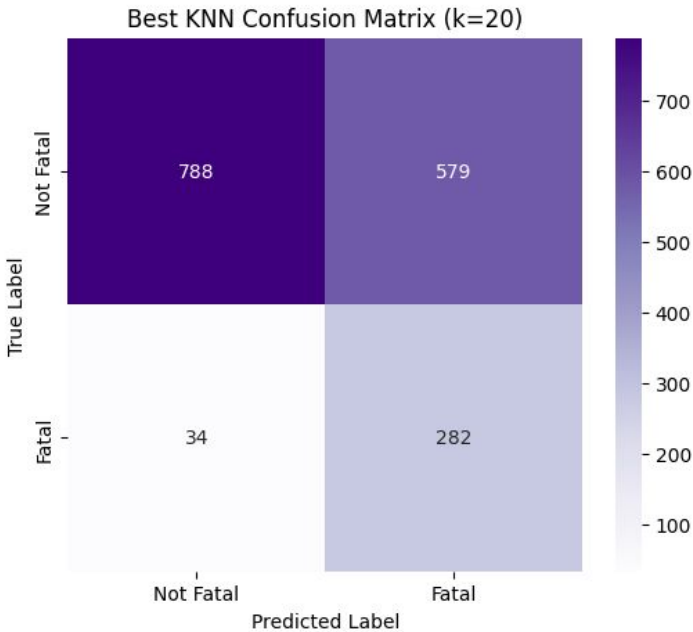
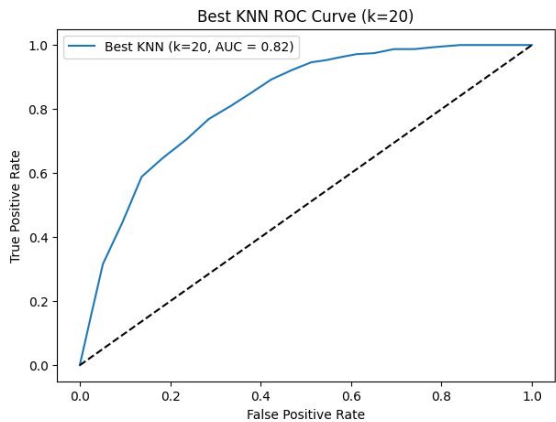
Decision Tree Visualization (Full Model, Pruned for Display)



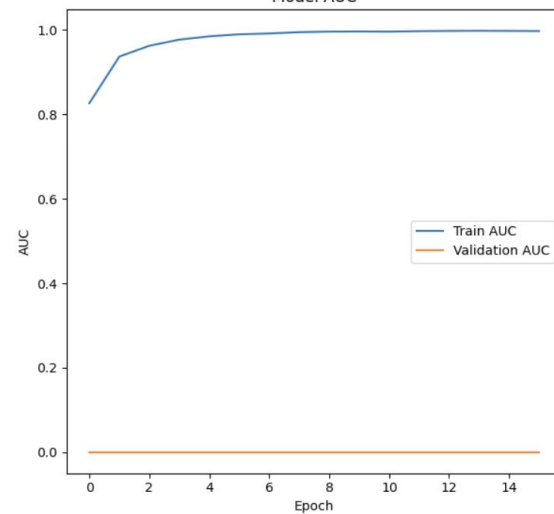
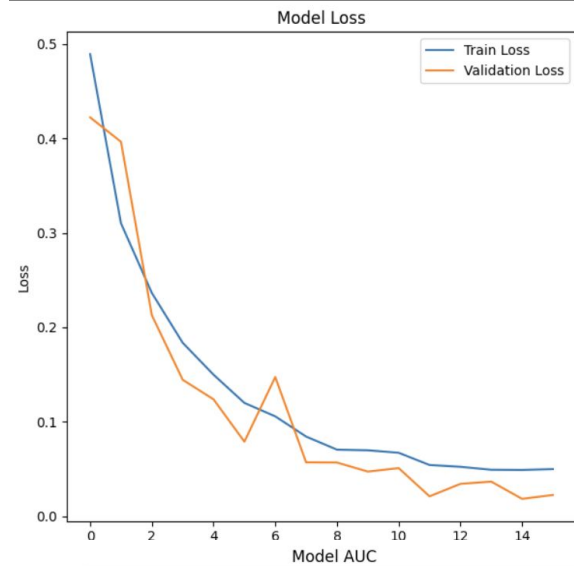
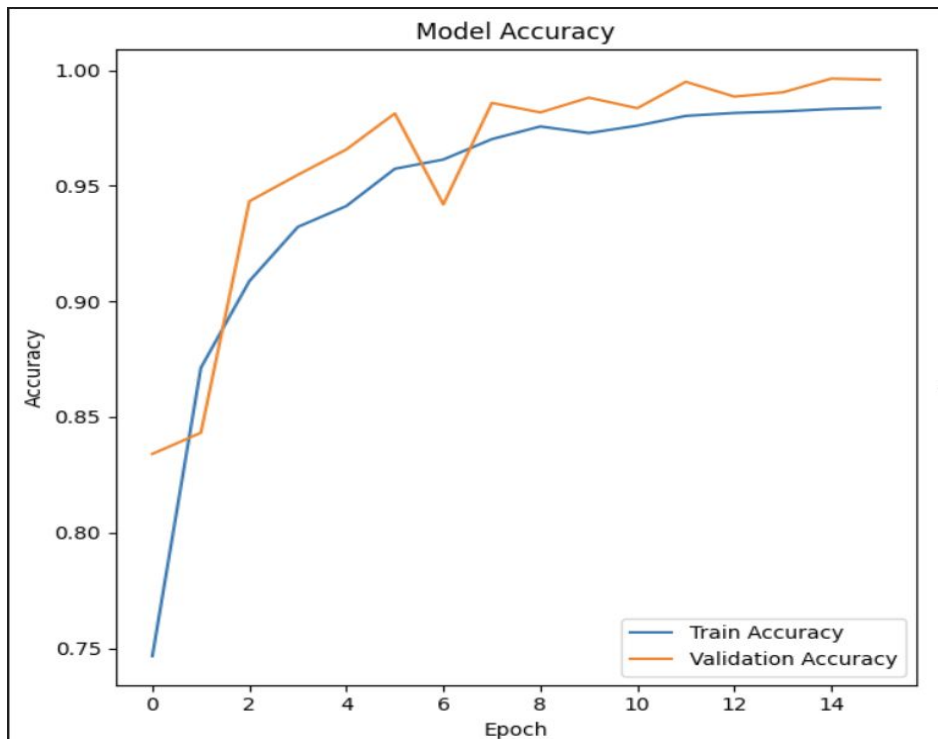
Model: KNN

Classification Report for k = 20:

	precision	recall	f1-score	support
Not Fatal (0)	0.96	0.58	0.72	1367
Fatal (1)	0.33	0.89	0.48	316
accuracy			0.64	1683
macro avg	0.64	0.73	0.60	1683
weighted avg	0.84	0.64	0.67	1683



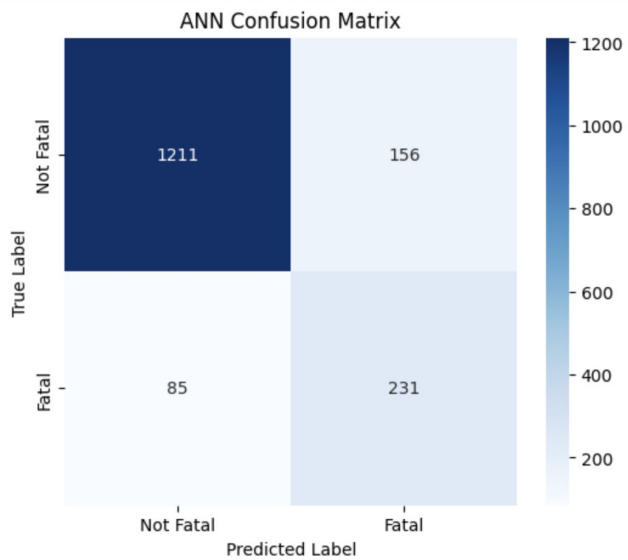
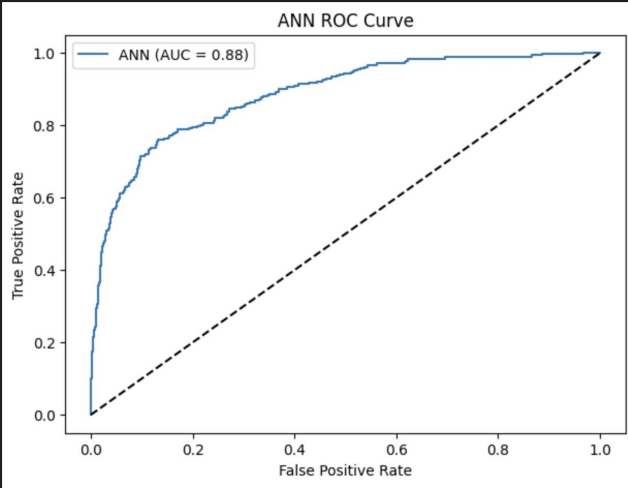
Model: ANN



Model: ANN

ANN Report:

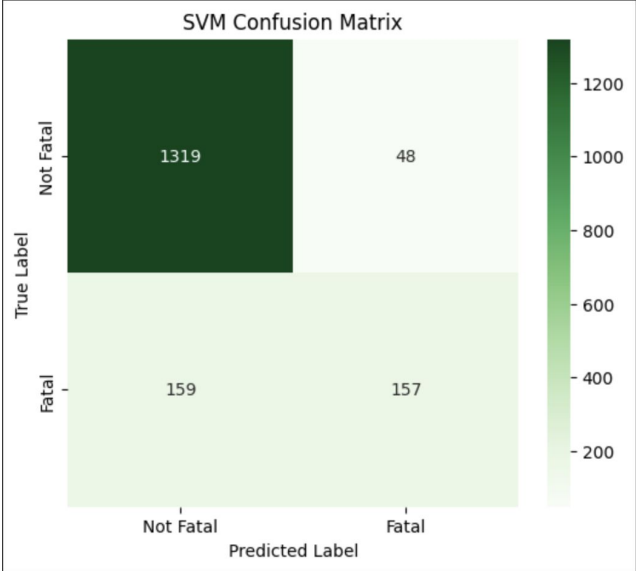
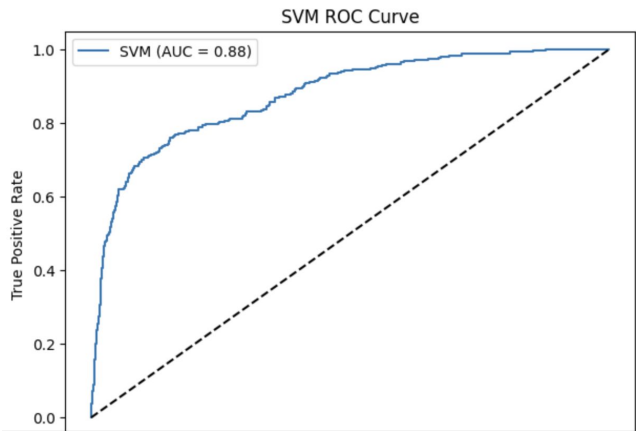
	precision	recall	f1-score	support
Not Fatal (0)	0.93	0.89	0.91	1367
Fatal (1)	0.60	0.73	0.66	316
accuracy			0.86	1683
macro avg	0.77	0.81	0.78	1683
weighted avg	0.87	0.86	0.86	1683



Model: SVM

ANN Report:

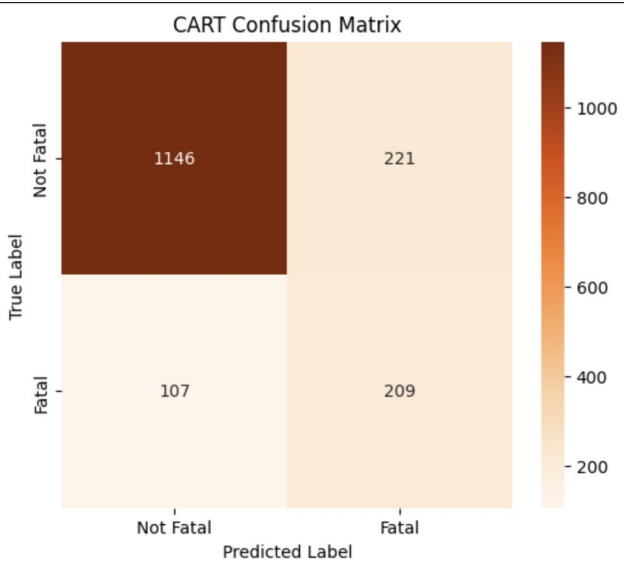
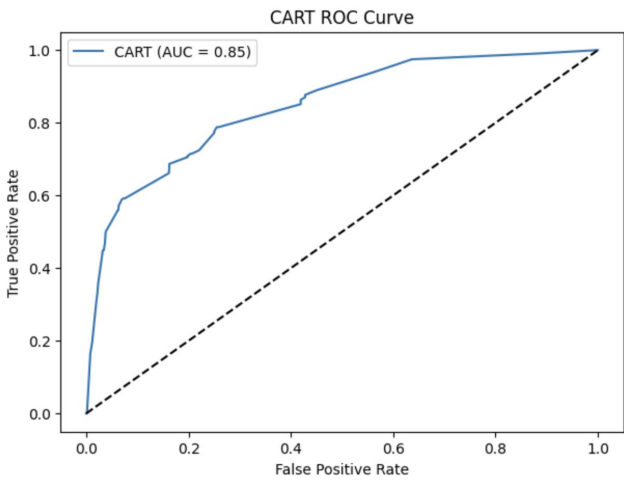
	precision	recall	f1-score	support
Not Fatal (0)	0.89	0.96	0.93	1367
Fatal (1)	0.77	0.50	0.60	316
accuracy			0.88	1683
macro avg	0.83	0.73	0.76	1683
weighted avg	0.87	0.88	0.87	1683



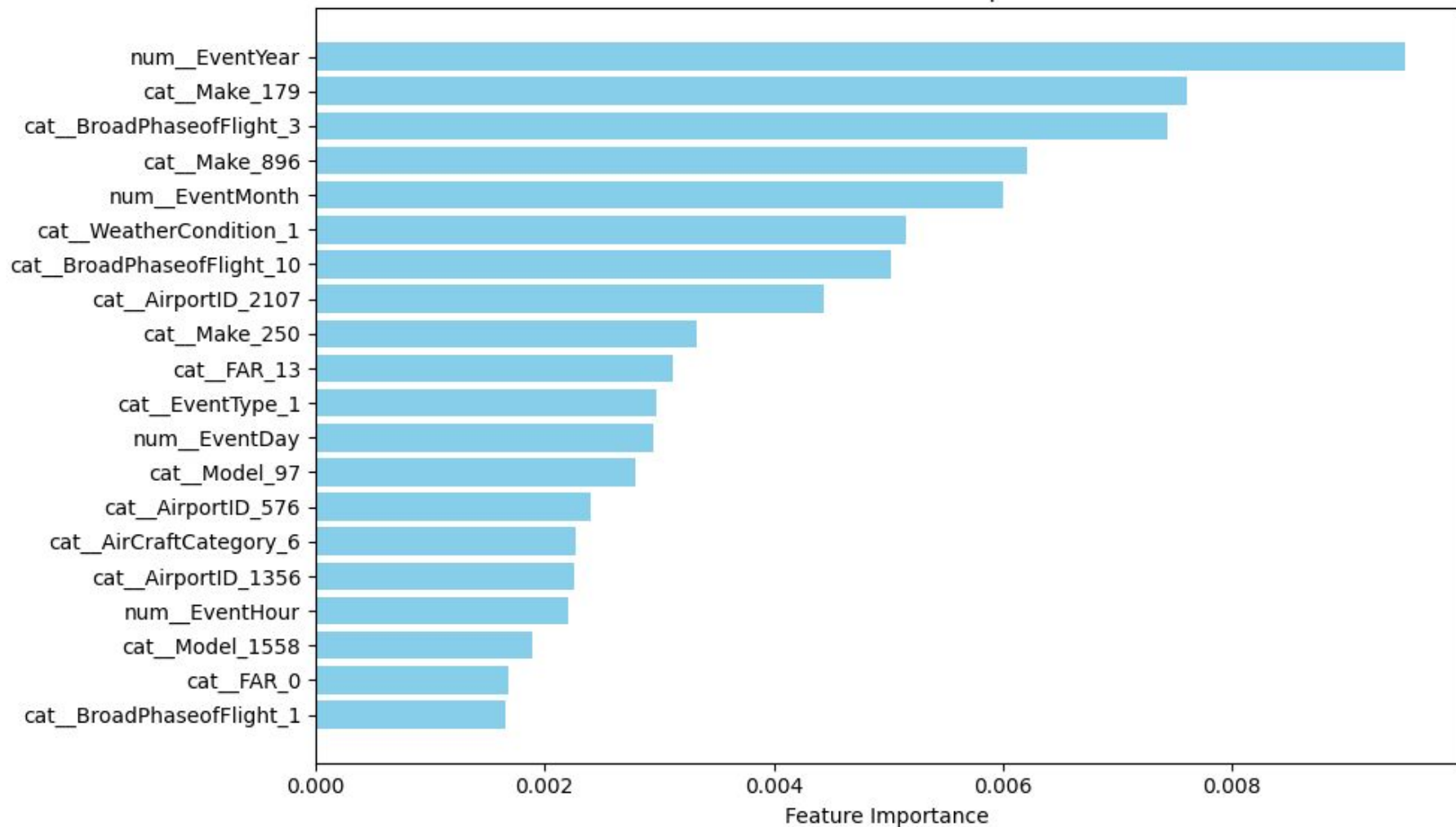
Model: CART

ANN Report:

	precision	recall	f1-score	support
Not Fatal (0)	0.91	0.84	0.87	1367
Fatal (1)	0.49	0.66	0.56	316
accuracy			0.81	1683
macro avg	0.70	0.75	0.72	1683
weighted avg	0.83	0.81	0.82	1683



Random Forest Feature Importances



Conclusion

Key Takeaways

Objective:

- We aimed to predict the likelihood of fatal outcomes in aviation accidents using machine learning techniques.
- Key predictors of fatal outcomes include factors like aircraft type, phase of flight, and location.

Model Performance:

- Random Forest and ANN models achieved high AUC scores, showing strong predictive power.
-

Future Directions

Model Enhancements

- Incorporate temporal trends (e.g., seasonality or changes in safety standards over time).
- Use ensemble blending of models (e.g., combining ANN and Random Forest) for improved accuracy.
- Explore explainable AI (XAI) tools like SHAP to enhance interpretability

Data Improvements

- Integrate additional datasets (e.g., weather conditions, maintenance history, ATC communication).
- Enhance geographic information for location-based risk analysis.

Real-World Applications

- Develop a risk scoring system for new accidents as they're reported.
- Create a dashboard for aviation regulators or safety teams to monitor high-risk indicators in real-time.